

Bilal Amin

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EDUCATION

Universität des Saarlandes <i>Master's degree, Data Science and AI</i> <i>Grade: 1.9 (German grading system)</i>	Apr. 2024 – Present Saarbrücken, Germany
University of Bergen (Erasmus+ Exchange Semester) <i>Exchange Student, Computer Science</i>	Jan. 2026 – Jun. 2026 Bergen, Norway
National University of Sciences & Technology <i>Bachelor's in Mechatronics Engineering (Computer Science)</i> <i>Grade: 1.7 (German grading system)</i>	Sep. 2017 – Sep. 2021 Islamabad, Pakistan

EXPERIENCE

Work Student – Internet Architecture Group <i>Max Planck Institute for Informatics (MPI)</i> - Designed an automated framework to evaluate congestion control algorithms in NS-3 (Network Simulator). - Created a user-friendly dashboard to visualize performance metrics under different network conditions. - Developed system tests and metrics for performance evaluation across varied scenarios. - Analyzed simulation results to identify optimization opportunities and guide design decisions.	Oct. 2024 – Mar. 2025 Saarbrücken, Germany
Data Analyst <i>Convergent Business Technologies</i> - Developed reproducible data science workflows using Python and SQL to manipulate and analyze data. - AS part of training took and finished the CS50 course and made several dashboards regarding it.	October. 2023 – March. 2024 Islamabad, Pakistan
Web Development Intern <i>CureMD</i> - Designed user interfaces using HTML, CSS, and JavaScript, improving usability and accessibility for end users. - Developed APIs in the .NET framework, enhancing data integration and tool development.	June 2023 – August 2023 Lahore, Pakistan
Technical Intern <i>STEP (Space Technology Education and Popularization)</i> - Led workshops teaching programming fundamentals using Scratch, focusing on hands-on learning for underprivileged youth. - Created educational materials to simplify complex technology concepts, fostering interest in STEM fields.	August 2021 - October 2021 Islamabad, Pakistan

PROJECTS

GermanLyft – AI-Powered German Learning Platform - Developed a full-stack German-learning platform using React, TypeScript, Flask, and the Groq LLM API, targeting primarily CEFR A1–A2 learners. - Built an interactive AI tutor for German grammar explanations, sentence correction, vocabulary practice, conversational learning, and dynamically generated exercises. - Implemented placement tests, quizzes, user accounts, profile functionality, and Anki .apkg deck importing for flashcard-based learning. - Designed a REST-based architecture with a Vite/React frontend and lightweight Flask backend, with production deployment using Vercel and Render.	2026
Smart StudyDesk: AI-Powered Study Organizer - Designed a cross-platform study application for organizing, summarizing, and reviewing lecture materials. - Added AI-based note summaries, personalized quizzes, progress tracking, and weak-area analysis. - Conducted usability testing using semi-structured interviews and NASA-TLX workload analysis. - Improved the interface based on user feedback, with clearer navigation, folder organization, and feedback visibility.	2025
AI Learning Assistant using LLMs (RAG)	2025

- Built an AI-powered learning assistant using Python, LangChain, Streamlit, and FastAPI to support document-based learning.
- Integrated a locally hosted Llama model with Retrieval-Augmented Generation (RAG) to answer questions from uploaded PDF documents.
- Implemented document processing, vector embeddings, ChromaDB retrieval, and citation-based responses for grounded question answering.
- Developed automatic generation of summaries, quizzes, and study materials from uploaded course documents.

MarketPulse Website Design and HCI Analysis

2024

- Designed and developed a responsive website for MarketPulse using HTML, CSS, and JavaScript.
- Applied HCI principles including consistency, simplicity, feedback, visibility, constraints, and accessibility.
- Improved the information structure and user interaction through usability-focused design.

Side Hustle Launchpad – Tech4Europe Hackathon, KIT

2025

- Developed a full-stack web application during the Tech4Europe Hackathon at Karlsruhe Institute of Technology, Germany.
- Built analytics dashboards and personalized recommendations to support small-business planning.
- Used FastAPI and MongoDB for backend development and data management.

MOOCS AND RELEVANT COURSEWORK

- **Saarland University:** Data Networks, Neural Networks: Theory and Implementation, Elements of Machine Learning, High-Level Computer Vision, Generative AI, Attacks Against Machine Learning Models, and Competitive Programming.
- **University of Bergen, 2026:** Qualitative Analysis and Functional Programming.
- **Machine Learning Specialization** — DeepLearning.AI and Stanford Online.
- **CS50's Introduction to Artificial Intelligence with Python** — Harvard University.
- **Python for Data Science, AI & Development** — IBM.
- **Google Data Analytics Professional Certificate** — Google.

TECHNICAL SKILLS AND INTERESTS

Programming: Python, JavaScript, TypeScript, SQL, HTML, CSS, C++, C#, Java, Dart

Machine Learning & AI: Scikit-Learn, Large Language Models (LLMs), Prompt Engineering, Retrieval-Augmented Generation (RAG)

LLM Tools & APIs: Groq API, LangChain, Ollama, ChromaDB

Web & App Development: Flask, FastAPI, ReactJS, Vite, Streamlit, Flutter, REST APIs

Databases: SQLite, Supabase, MongoDB, MySQL

Data Science Libraries: Pandas, NumPy, Matplotlib

Deployment & Dev Tools: Vercel, Render, Docker, Git, GitHub, GitHub Actions, VS Code, Jupyter

Visualization & Design: Tableau, Figma, Canva

Interests: Artificial Intelligence, LLM Applications, Generative AI, Intelligent Learning Systems, Machine Learning, Data Visualization